

Sales Prediction Report

Multiple Linear Regression

Yuanzheng Zheng | DATA 555 Project 1

I used multiple linear regression to predict Sales from TV, Radio, Social Media, and Influencer type. The original dataset has 4,572 rows and five columns. I removed 26 rows with missing values, leaving 4,546 complete rows with no duplicates.

TV and Sales follow a strong, nearly straight upward pattern, while Radio and Social Media show more spread. I changed Influencer into three 0/1 columns, with Macro as the reference category. I used 3,636 rows for training and 910 rows for testing.

On the test set, Mean Absolute Error was 2.3129, Root Mean Squared Error was 2.8849, and R-squared was 0.9990. The predicted values were close to actual Sales, and the residual plot had points on both sides of zero without a clear curved pattern. TV had a coefficient of about 3.56, meaning one more unit of TV was linked to about 3.56 more units of predicted Sales when the other inputs stayed the same. The other coefficients were close to zero, but this dummy dataset does not prove that advertising causes higher Sales or that the model will work as well with real data.

AI Use Record

I used ChatGPT for advice on Influencer encoding and Markdown layout. I also used the Codex assistant to check the notebook and draft the proposal and report.

I asked these two questions in a later follow-up. The answers below were generated then; they are not the earlier ChatGPT replies mentioned in the notebook.

Prompt	Output
1. how to handle data in categorical data in machine learning	Categorical data contains labels, such as Macro, Mega, Micro, and Nano. Many machine learning models need numbers as inputs. Use one-hot encoding for categories with no order: it creates columns with values of 0 or 1. For categories with a real order, such as low, medium, and high, ordinal encoding may be suitable. For linear regression with an intercept, drop one dummy column and use that category as the reference. For your notebook, use the following code; Macro is the reference category, and Sales should remain the target rather than an input. <pre>model_data = pd.get_dummies(marketing_data, columns=['Influencer'], drop_first=True, dtype=int)</pre>
Prompt	Output
2. what's the basic syntax of Markdown, simple use	Markdown is a simple way to format text. In a notebook, use a Markdown cell for these examples: <pre># Title ## Section **bold** and *italic* - List item 1. Numbered item [Link](https://example.com) `column_name`</pre> These show headings, bold and italic text, lists, a link, and inline code. Leave a blank line between paragraphs.

Help with these documents

Actual writing request in English translation Please write the proposal and summary report based on my notebook, I'm not fully in speaking english.

Output The proposal text on page 1 is the draft produced from this request. The assistant also prepared this AI use record.

Validation

The notebook lists the four Influencer categories and shows three encoded columns. The displayed Macro row has zero in all three columns. The encoding keeps all 4,546 complete rows. The code keeps Sales out of X and uses it as y. During the AI-assisted review, the assistant ran all 13 code cells using the repository data. The run returned

Mean Absolute Error 2.3129, Root Mean Squared Error 2.8849, and R-squared 0.9990, matching the notebook text. The assistant also checked that the proposal and report use the same data counts, methods, and results.

Data source

Harriman Samuel Saragih. Dummy Marketing and Sales Data. Kaggle.

[Dummy Marketing and Sales Data on Kaggle](#)